EE 152-BASIC ELECTRONICS

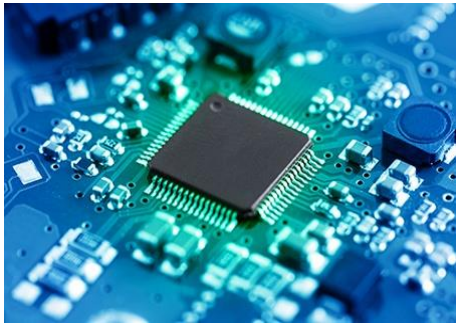
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CHAPTER ONE

SEMI-CONDUCTOR MATERIALS AND PROPERTIES

CONDUCTORS, INSULATORS AND SEMI-CONDUCTORS







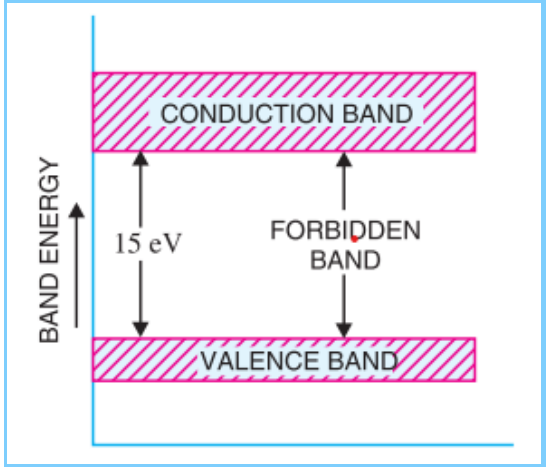
Electronic Materials

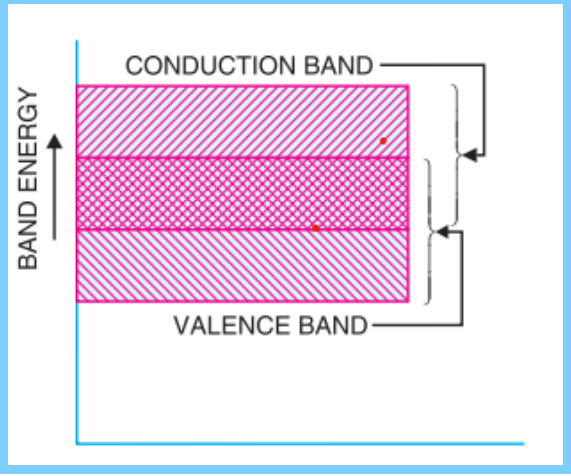
The goal of electronic materials is to generate and control the flow of an electrical current. Electronic materials can be classified into the following:

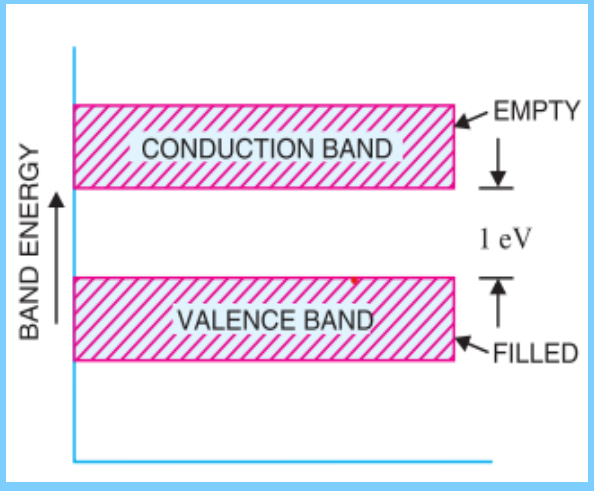
- **Conductors: Have low resistance which allows electrical current flow.**
- **Insulators: have high resistance which suppresses electrical current flow.**
- **Semiconductors: Can allow or suppress electrical current flow.**

Conductors, Semiconductors and Insulators

The difference in the behaviour of electronic materials in terms of their conductivity can be explained with the help of energy bands.







Insulators
Large energy difference between conduction band and valence band.

Conductors
Conduction band and valence band overlap.

Semi – Conductors
Small energy difference between conduction band and valence band.

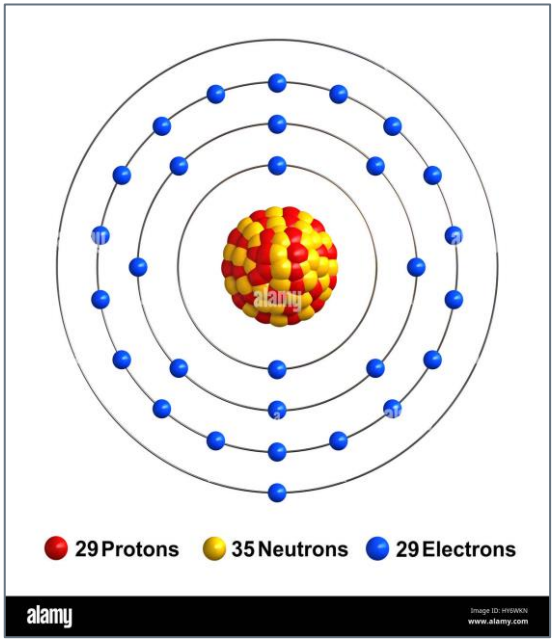
Conductors

Good conductors have low resistance so electrons flow through them with ease.

- **Best element conductors include: Copper, silver, gold, aluminium, & nickel.**
- **Alloys are also good conductors: Brass & steel**
- **Good conductors can also be liquid: Salt water**

Atomic Structure of Conductors

- The atomic structure of good conductors usually includes only one electron in their outer shell.
- It is called a valence electron.
- It is easily striped from the atom, producing current flow.



Copper Atom

Insulators

- Insulators have a high resistance so current does not flow in them.
- Good insulators include: Glass, ceramic, plastics, & wood.
- Most insulators are compounds of several elements.
- The atoms are tightly bound to one another so electrons are difficult to strip away for current flow.

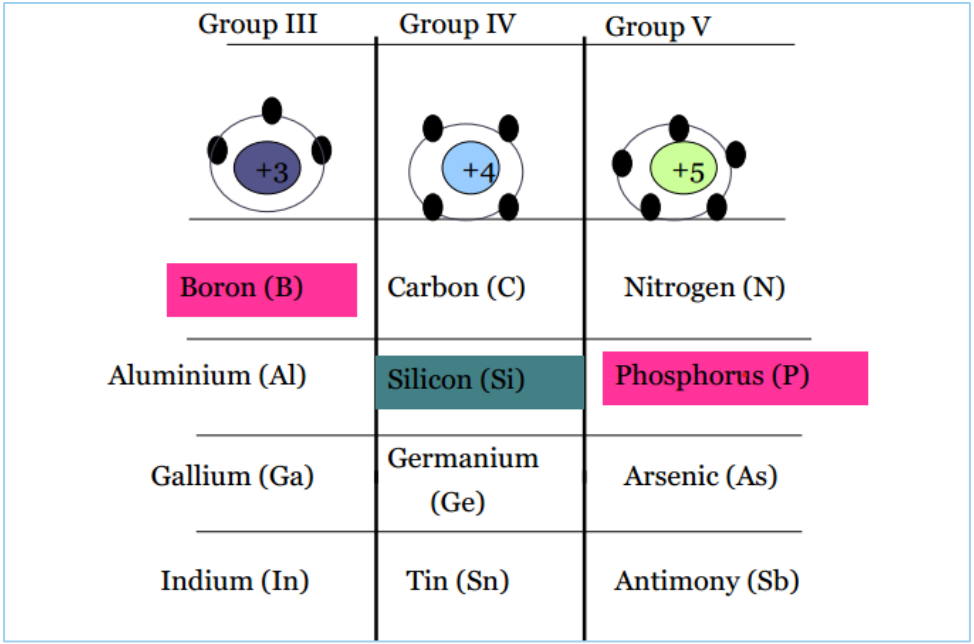
Semi-Conductors

- A semiconductor is a material with conducting properties between those of a good insulator and a good conductor.
- Common elements such as carbon, silicon, and germanium are semiconductors.
- Silicon is the best and most widely used semiconductor.

Questions

1. Why is Silicon the most widely used semiconductor?
2. Why is Silicon preferred to Germanium?

Semi-Conductors Elements in the Periodic Table

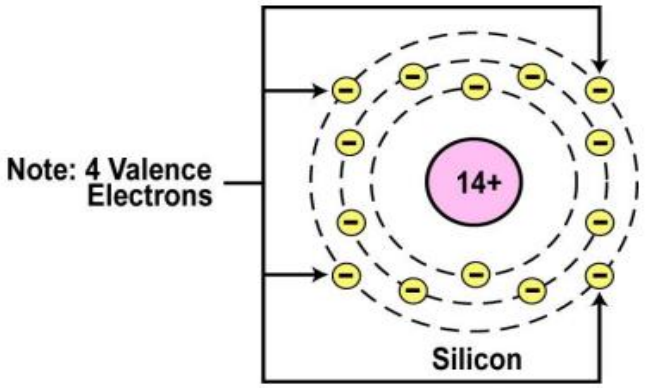


Question

- **Why are Group 2 or Group 6 elements not used to dope semiconductors?**

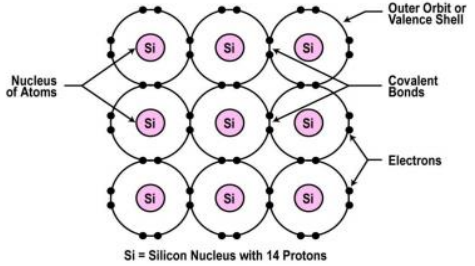
Semi-Conductor Valence Orbit

The main characteristic of semiconductor element is that it has four electrons in its outer or valence orbit.

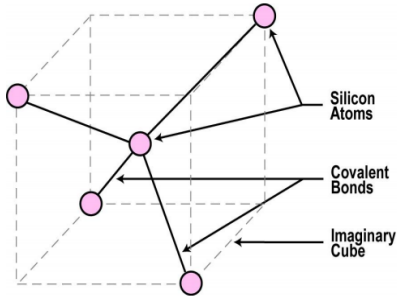


Crystal Lattice Structure

- A crystal is a substance whose atoms or molecules are arranged in a ordered pattern.
- The unique capability of semiconductor atoms is their ability to link together to form a physical structure called a crystal lattice.
- The atoms link together with one another sharing their outer electrons.
- These links are called covalent bonds.



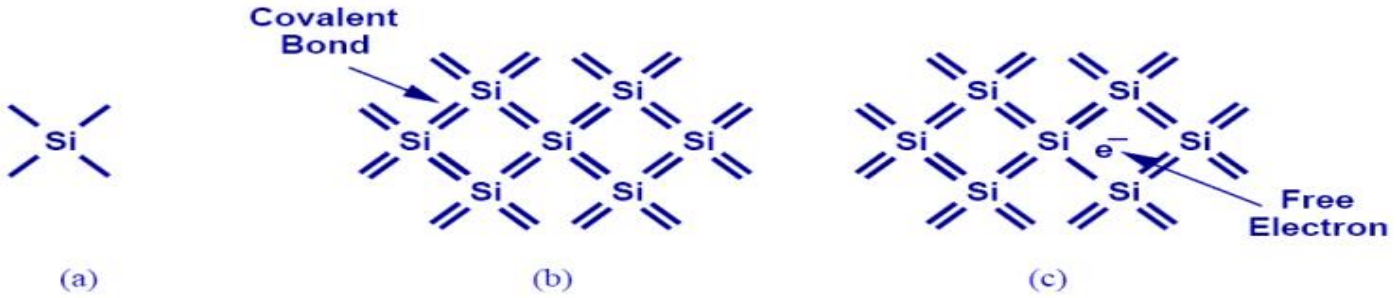
2-D Structure



3-D Structure

Silicon

- Silicon is a tetravalent element, which means it has four valence electrons and four vacancies.
- In intrinsic (pure) silicon, atoms join together by forming covalent bonds. Each atom shares its valence electrons with each of four adjacent neighbours effectively filling its outer shell.
- When temperature goes up, electrons can become free to move about the Si lattice.
- Atomic density: 5×10^{22} atoms/cm^3



Electronic Properties of Silicon

- **Silicon is a semiconductor material.**
- **Pure Si has a relatively high electrical resistivity at room temperature.**

There are two(2) types of mobile charge-carriers in Si:

- **Electrons - negatively charged**
- **Holes - are positively charged**

Electronic Properties of Silicon

The concentration (#/cm^3) of conduction electrons & holes in a semiconductor can be modulated in several ways:

- **by adding special impurity atoms (dopants)**
- **by applying an electric field**
- **by changing the temperature**
- **by irradiation**

Classification of Semiconductors

Semiconductors are classified into two types:

- **Intrinsic semiconductors**
- **Extrinsic semiconductors**

Intrinsic Semiconductors

- **A semiconductor is said to be intrinsic if it contains no impurities and no crystalline defects.**

- **The structure has zero overall charge.**

Intrinsic Semiconductors: Behaviour at Absolute Zero

- At absolute zero, every electron is at the lowest energy state.
- Electrons occupy all of the electronic states in the valence band, and the states in the conduction band are empty.
- The intrinsic semiconductor thus behaves as a perfect insulator at absolute zero temperature.
- However, the covalent bonds can be broken if sufficient energy is applied to the semiconductor.

Intrinsic Semiconductors: Behaviour at room or higher temperatures

- At room or higher temperatures, electrons acquire thermal energy, which is transferred to them from the crystal lattice.
- The atoms in the crystal lattice vibrate, and these vibrations can be transmitted through the crystal as acoustic waves called **phonons**.
- Phonons can excite electrons from the valence band (VB) to the conduction band (CB).
- Electrons leaving the VB to CB creates a vacancy called **hole in the valence band**. A hole is a vacancy in a covalent bond with a unit positive charge associated with it.

Intrinsic Semiconductors: Behaviour at room or higher temperatures

- The electron is now free, and there is a free hole in the valence band.
- Every electron that escapes to conduction band creates a corresponding hole in the valence band, creating an electron-hole pair.
- When electron-hole pair is created by absorption of phonons, we call the process thermal generation.
- If photons provide the energy, the process is termed optical generation.

Intrinsic Semiconductors

- The free electrons in the semiconductor can fall into holes in a process known as recombination.
- Energy is given up by the electrons in this process.
- An intrinsic semiconductor is said to be in thermal equilibrium at any given temperature when the recombination rate is equal to the ionization rate.
- For an intrinsic semiconductor at thermal equilibrium, the equilibrium concentration n_o of electrons in the conduction band is the same as the equilibrium concentration p_o of holes in the valence band:

$$n_o = p_o = n_i$$

Where n_i is the equilibrium carrier concentration in an intrinsic semiconductor

Law of mass action

For an intrinsic semiconductor

$$n_o = p_o = n_i$$

n = concentration of electrons per unit volume

p = concentration of holes per unit volume

- **This is due to fact that when an electron makes a transition to the conduction band, it leaves a hole behind in valence band .**
- **Thus the number of holes and electrons are equal, hence**

$$np = n_i^2$$

- **This equation is called as mass-action law.**
- **This equation is valid for extrinsic as well.**

Carrier Concentration in Intrinsic Si

The intrinsic carrier concentration n_i depends on;

- **the semiconductor material, and**
- **the temperature**

For silicon at 300 K, n_i has a value of $1.5 \times 10^{10} cm^{-3}$

Doping

- To increase the conductivity of an intrinsic semiconductor, one can introduce an impurity in a process known as doping.

- These impurities are atoms from column III or V of the periodic table. They are referred to as dopants.

Extrinsic Semiconductors

- **Intrinsic conduction is very small.**
- **Conductivity levels can be raised and controlled by doping with minute levels of impurity atoms to give extrinsic or doped semiconductors.**

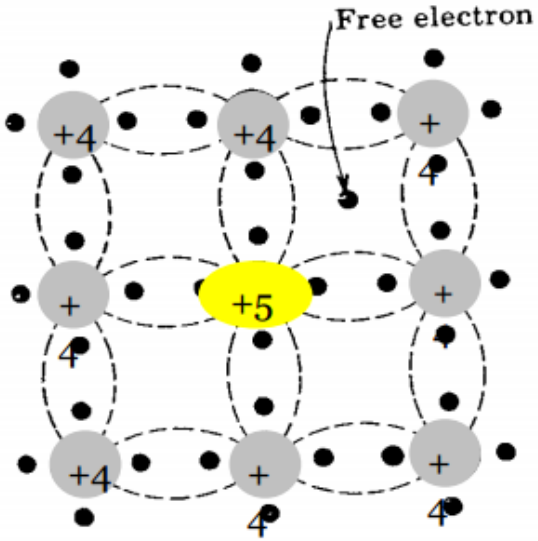
Extrinsic semiconductors may be further divided into either

- **N-type**
- **P-type**

N-type Semiconductors

- This is formed by adding a pentavalent atom to the intrinsic semiconductor.
- These atoms with five valence electrons are known as donor atoms because they provide an extra electron to the semiconductor's crystal structure.
- The extra electron becomes a conduction electron.

N-type Semiconductors



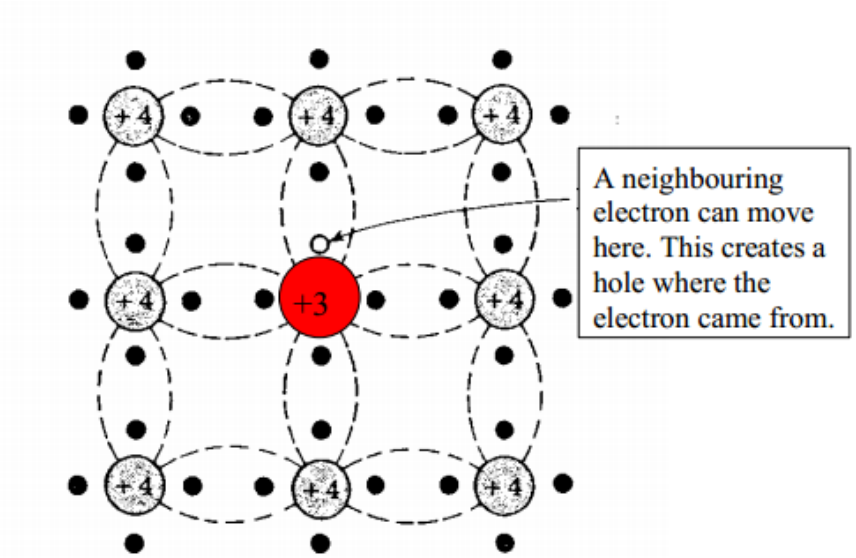
For N-type, n >> p;

- n is the majority carrier concentration n_n
- p is the minority carrier concentration p_n

P-type Semiconductors

- Here the doping atom has only three electrons in its outer shell.
- It is relatively easy for an electron from a neighbouring atom to move in, so releasing a hole at its parent atom.
- The freed hole is available for conduction.
- The energy needed to free the electron from its parent is usually small compared to the thermal energy so each impurity atom contributes one hole for conduction (fully ionised).

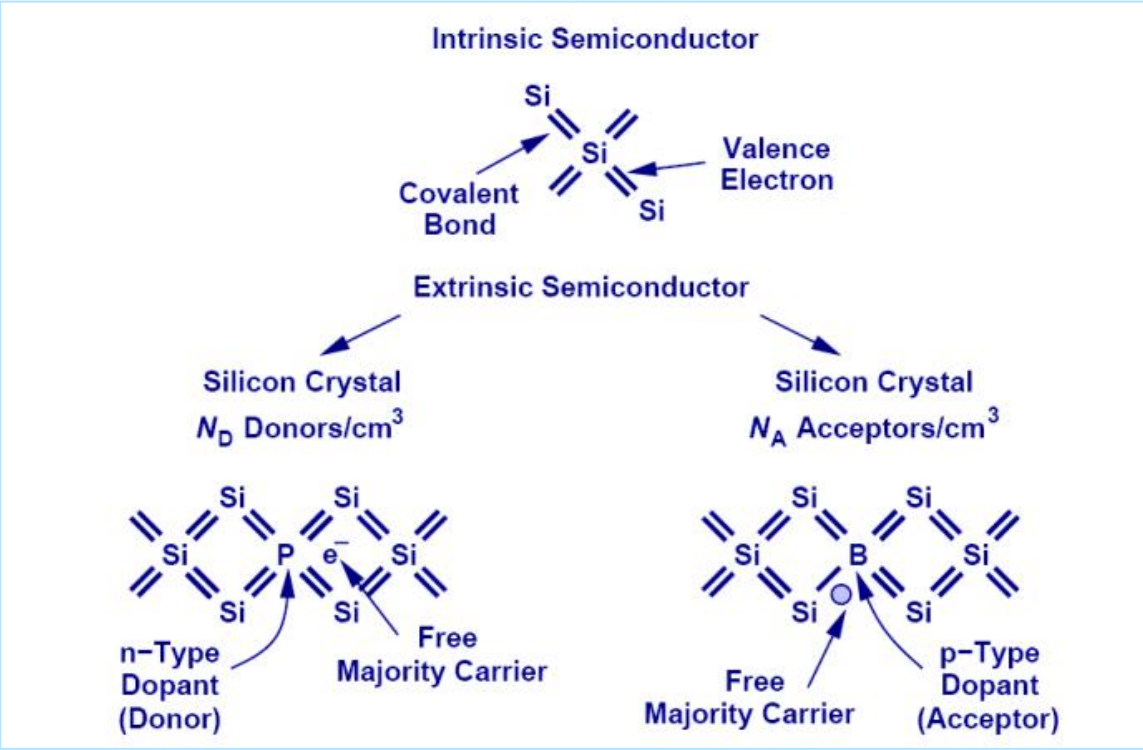
P-type Semiconductors



For P-type, p >> n;

- **p is the majority carrier concentration p_p**
- **n is the minority carrier concentration n_p**

Summary of Charge Carriers



Electron and Hole Concentrations

Under thermal equilibrium conditions, the product of the conduction-electron density and the hole density is ALWAYS equal to the square of n_i: $np = ni^2$

N-type material

$$n \approx N_D$$

$$p \approx \frac{n_i^2}{N_D}$$

$N_D = donor\ concentration$

P-type material

$$p \approx N_A$$

$$n \approx \frac{n_i^2}{N_A}$$

$N_A = acceptor\ concentration$

Example 1

Calculate the hole and electron densities in a piece of p-type silicon that has been doped with 5×10^{16} acceptor atoms per cm^3. $n_i = 1.5 \times 10^{10}\ cm^3$ (at room temperature).

Solution

P-type ; $p \gg n$

$np = n_i^2$

$N_A = p = 5 \times 10^{16}\ cm^{-3}$

$$n = \frac{n_i^2}{p} = \frac{(1.5 \times 1010)^2}{5 \times 1016}$$

$n = 4.5 \times 10^3$ electrons per cm^3

NB: $p \gg n_i$ and $n \ll n_i$ in a p-type material. The more holes you put in the less electrons you have and vice versa

Carrier Movement



- **Drift Current**
- **Diffusion Current**

Carrier Movement

There are two mechanisms by which holes and free electrons move through a silicon crystal.

Drift

The carrier motion is generated by the electrical field across a piece of silicon. This motion will produce drift current.

Diffusion

The carrier motion is generated by the different concentration of carrier in a piece of silicon. The diffused motion, usually carriers diffuse from high concentration to low concentration, will give rise to diffusion current.

Carrier Mobility

Mobility

- Mobility is a measure of how easily a carrier moves in particular material.
- It is normally expressed in centimetres squared per volt-second (cm^2/Vs).
- Carrier mobility, $\mu = \frac{v}{E}$ and it is the drift velocity per unit field.

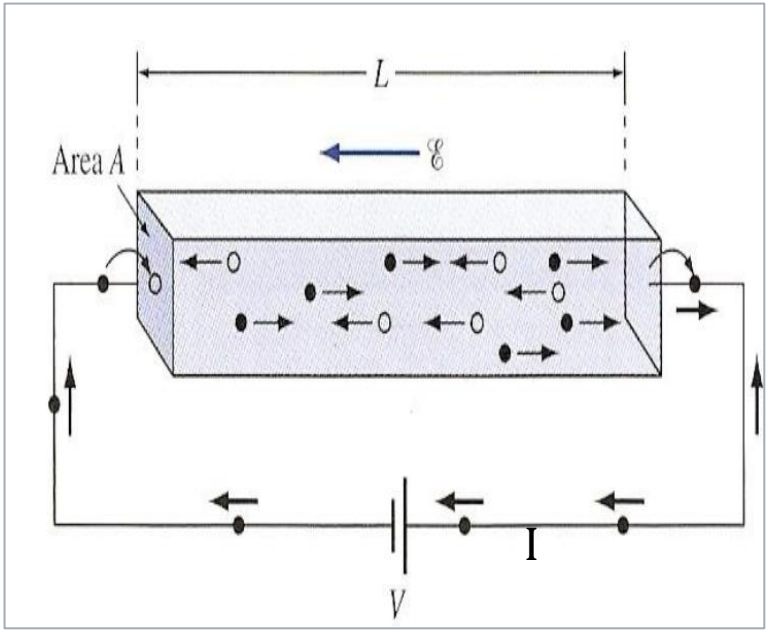
Drift velocity

- Drift velocity is the average velocity with which the charge carriers drift in a definite direction under the influence of an applied electric field.
- The unit of μ is $\frac{cm/s}{V/cm} = cm^2/Vs$

Drift Current

- Drift current is produced by the motion of electrons and holes under an electric field.
- The net current is always in the direction of applied electric field.
- The current due to both charges move in the same direction.

$$I_{drift} = I_{n(drift)} + I_{p(drift)}$$



Current Density

- In a semiconductor, it is often more useful to talk about current density.
- Current Density (J) is the amount of charge crossing a plane unit area per unit time.

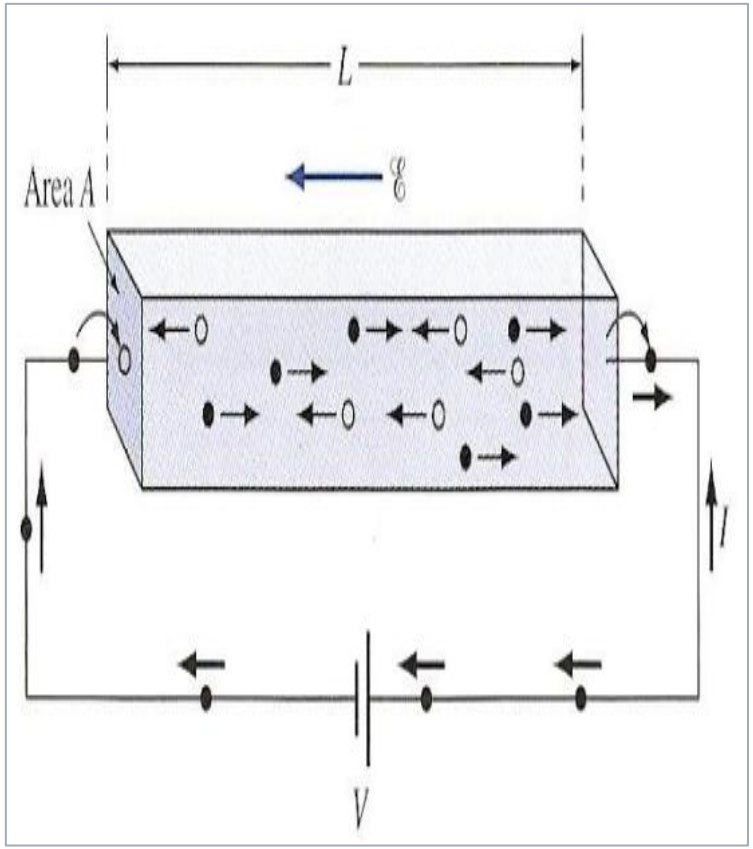
$$J = \frac{I}{A}$$

- For a semiconductor with an applied electric field, the total drift current density is:

$$J_{drift} = J_{n(drift)} + J_{p(drift)}$$

Where

- $J_{n(drift)}$ is the current density due to electrons
- $J_{p(drift)}$ is the current density due to holes



Conductivity

- When an electric field E is applied to a semiconductor, the free electrons drift with average velocity;

$$v_e = -\mu_e E$$, where μ_e is the electron mobility.

- The holes drift with a velocity;

$$v_h = +\mu_h E$$, where μ_h is the hole mobility.

- Although the free electrons and holes drift in opposite directions, the current densities add because the charge polarities are opposite.
- The electron charge density i $\rho_e = -nq$ and the hole charge density is $\rho_h = +pq$, where n and p represent the electron and hole concentrations respectively.

Conductivity

- The total conduction current density can be written as:

$$J_{drift} = \rho_e v_e + \rho_h v_h = (n\mu_e + p\mu_p)qE = \sigma E$$

- This equation defines the conductivity σ of the semiconductor. It is given by:

$$\sigma = (n\mu_e + p\mu_p)q$$

- Alternatively, the conductivity can be obtained from the dimensions of the semiconductor as follows.

Conductivity

- From ohm's law, the resistance R of a uniform sample of length L and cross-sectional area A as illustrated in the semiconductor block below, is

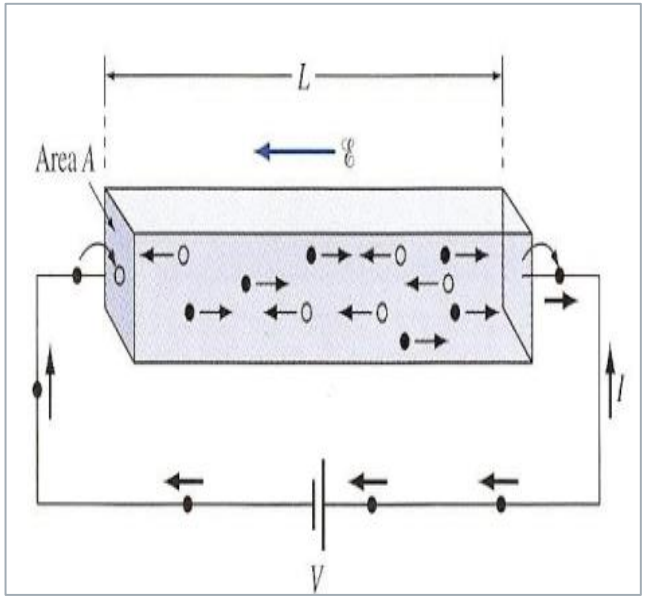
$$R = \frac{V}{I} = \frac{\rho L}{A}$$

- Since the semiconductor is uniform, V = EL and using I = JA,

$$\frac{EL}{JA} = \frac{\rho L}{A}$$

$$J_{drift} = \frac{E}{\rho} = \sigma E$$

- where $\sigma = \frac{1}{\rho}$ is the conductivity.
- E is the electric field and ρ is the resistivity



Example 2

A rod of intrinsic silicon is 1 cm long and has a diameter of 1mm. At room temperature, the intrinsic concentration in the silicon is $n_i = 1.5 \times 10^{16}$ per m^3. The electron and hole mobilities are $\mu_e = 0.13\ m^2V^{-1}\ s^{-1}$ and $\mu_h = 0.05\ m^2V^{-1}\ s^{-1}$. Calculate the conductivity of the silicon and the resistance R of the rod.

Solution

The conductivity is calculated as follows:

$$\sigma = n_i(\mu_e + \mu_h)q$$

$$\sigma = 1.5 \times 10^{16} \times (0.13 + 0.05) \times 1.602 \times 10^{-19}$$

$$\sigma = 4.33 \times 10^{-4}\ S/m$$

Solution (Conti…)

Resistance is calculated as:

$$R = \frac{L}{\sigma A} = \frac{0.01}{\pi \times (0.5 \times 10{-}3)^2 \times 4.33 \times 10{-}4}$$

R = 29.4 MΩ

Temperature sensitivity

- In both types of extrinsic semiconductor virtually all available charge carries are freed from their parent atoms at room temperature.
- Temperature variations thus make little difference to the conductivity, σ.
- For intrinsic conductivity the number of carriers, and thus, increases rapidly with temperature.
- For both extrinsic and intrinsic mechanisms the conductivity is zero at T = 0K

Diffusion Current

- The directional movement of charge carriers due to their concentration gradient produces a component of current known as diffusion current.
- It is encountered only in semiconductors and is normally absent in conductors.
- With no applied voltage, if the number of charge carriers (either holes or electrons) in one region of a semiconductor is less compared to the rest of the region, then there exists a concentration gradient.
- As a result, the carriers tend to move gradually or diffuse from the region of higher concentration to the region of lower concentration.

Diffusion Current

- This process is called diffusion and electric current produced due to this process is called diffusion current.
- This process continues until all the carriers are evenly distributed through the material.
- Hence when there is no applied voltage, the net diffusion current will be zero.

Diffusion Current

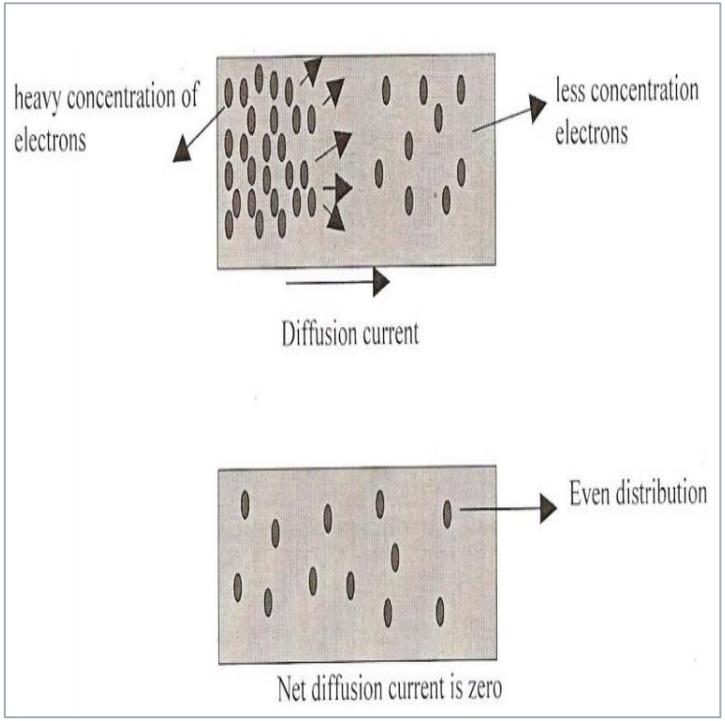
- In general, the hole diffusion current density is given by:

$$J_{p(diff)} = - qD_h \nabla_p$$

- Where D_h is the hole diffusion constant and ∇_p is the gradient of p.

- Similarly, the electron diffusion current density is given by

$$J_{n(diff)} = + qD_e \nabla_n$$



Total Current

The total current in a semiconductor = the electron and hole drift currents + the electron and hole diffusion currents.

$$J_{total} = J_{drift} + J_{diffusion}$$

$$J_{total} = J_{drift} + J_{p(diff)} + J_{n(diff)}$$

$$J_{total} = \sigma E + q(D_e \nabla_n - D_h \nabla_p)$$

Carrier Lifetime /Mean Lifetime

- **Because hole-electron pairs are continually created by thermal agitation of a semiconductor lattice, it might seem that the number of holes and free electrons would continually increase with time.**
- **This does not happen because free electrons are continually recombining with holes.**
- **At any temperature, a stable state is reached when the creation rate of hole-electron pairs is equal to the recombination rate.**

Carrier Lifetime /Mean Lifetime

- The mean lifetime τ_n of a free electron is the average time that the electron exists in the free state before recombination.
- The mean lifetime τ_p for the hole is defined similarly.
- In the intrinsic semiconductor, τ_n is equal to τ_p because the number of free electrons must be equal to the number of holes.

Terminologies

Donor

Impurity atom that increases n

Acceptor

Impurity atom that increases p

N-type material

Contains more electrons than holes

P-type material

Contains more holes than electrons

Majority carrier

The most abundant carrier

Minority carrier

the least abundant carrier

Intrinsic semiconductor

$n = p = n_i$

Extrinsic semiconductor

Doped semiconductor

Summary

- In its pure state, semiconductor material is an excellent insulator.
- The commonly used semiconductor material is silicon.
- Semiconductor materials can be doped with other atoms to add or subtract electrons.
- An N-type semiconductor material has extra electrons.
- In an N-type semiconductor, conduction is mainly due to electrons (negative charges).
- Positive charges (holes) are the minority carriers.

Summary

- A P-type semiconductor material has a shortage of electrons with vacancies called holes.
- In a P-type semiconductor, conduction is mainly due to holes (positive charges).
- Negative charges (electrons) are the minority carriers.
- The heavier the doping, the greater the conductivity or the lower the resistance.
- By controlling the doping of silicon the semiconductor material can be made as conductive as desired.

Thanks!

Any questions?